



Introduction to Big Data

Introduction to Big Data

- Big data is an all-inclusive term that refers to extremely large, very fast, highly diverse and complex data that cannot be managed with traditional data management tools
- It includes all kind of data, which helps deliver the right information, to right person, in right quantity, at right time, to make right decision
- Big data can be harnessed by developing infinitely scalable, flexible, and evolutionary data architectures, coupled with the use of cost-effective computing machines.

Definition

“Big data” is

*high-volume, -velocity and -variety information
assets*

*that demand cost-effective, innovative forms of
information processing*

for enhanced insight and decision making

By Gartner

Understanding Big Data

- Big data is data that exceeds the processing capacity of conventional database systems.
- The data is too big, moves too fast, or doesn't fit the structures of your database architectures.
- To gain value from this data, you must choose an alternative way to process it.
- At the fundamental level it is just another collection of data that can be analyzed and utilized for the benefit of the business and on another level, it is a special kind of data that poses unique challenges and offers unique benefits.

Life Cycle of Big Data

Big data is mostly, over 90%, unstructured data

There are huge opportunities for technology providers to innovate and manage the entire life cycle of Big Data

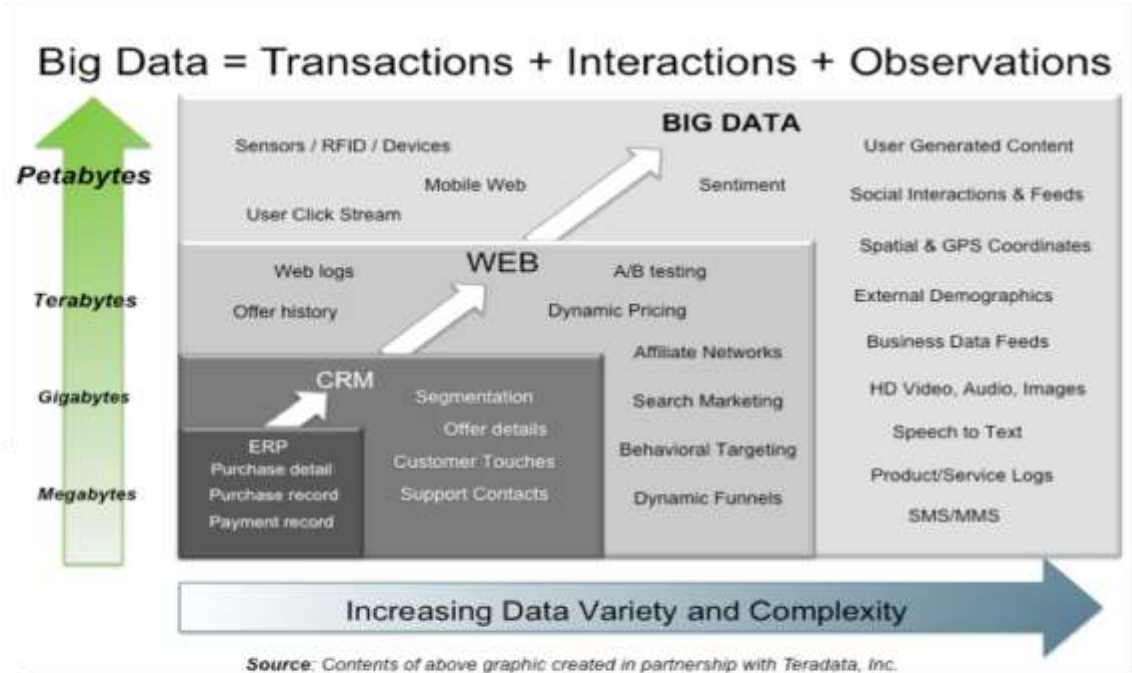
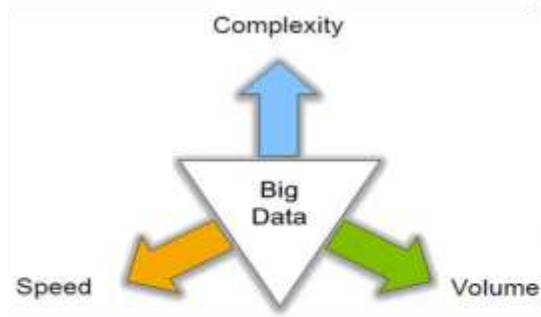
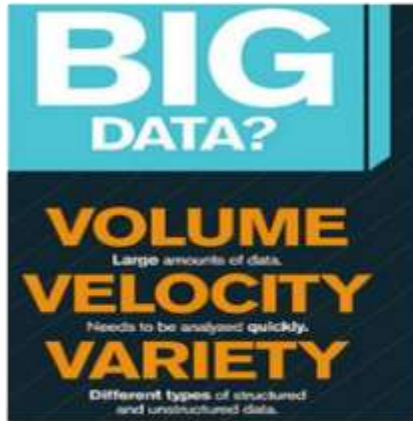
- Generate
- Gather
- Store
- Organize
- Analyze
- Visualize

The Three V-s

Volume and Velocity are driven by variety

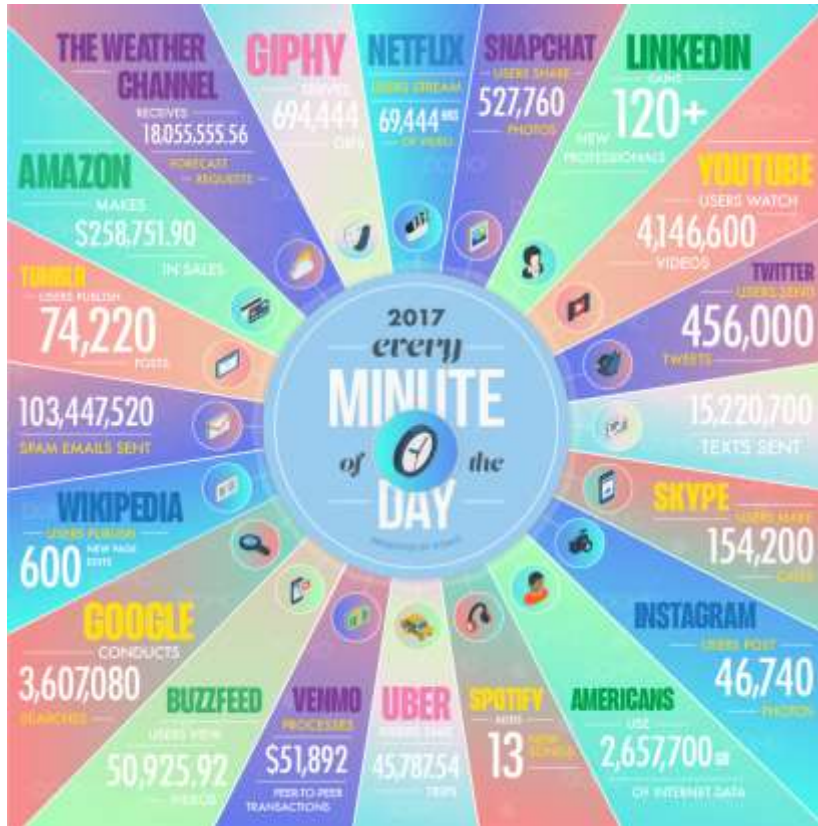
The varying varacity and value of data complicates the situation

Big Data: 3V's



Source: Contents of above graphic created in partnership with Teradata, Inc.

Volume



- Quantity of data generated in the world is doubling every 12-18 months
- Data sets too large to store and analyse using traditional databases and finding something from it in a reasonable period of time is like a miracle. (Petabytes and Exabytes) (1 Exabyte = 1 Million TB)

? TBs of
data every day

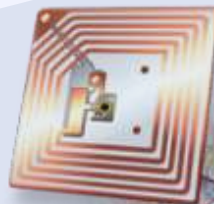


25+ TBs of
log data
every day



12+ TBs
of tweet data
every day

30 billion RFID
tags today
(1.3B in 2005)



76 million smart meters
in 2009...
200M by 2014



**4.6
billion**
camera
phones
world wide



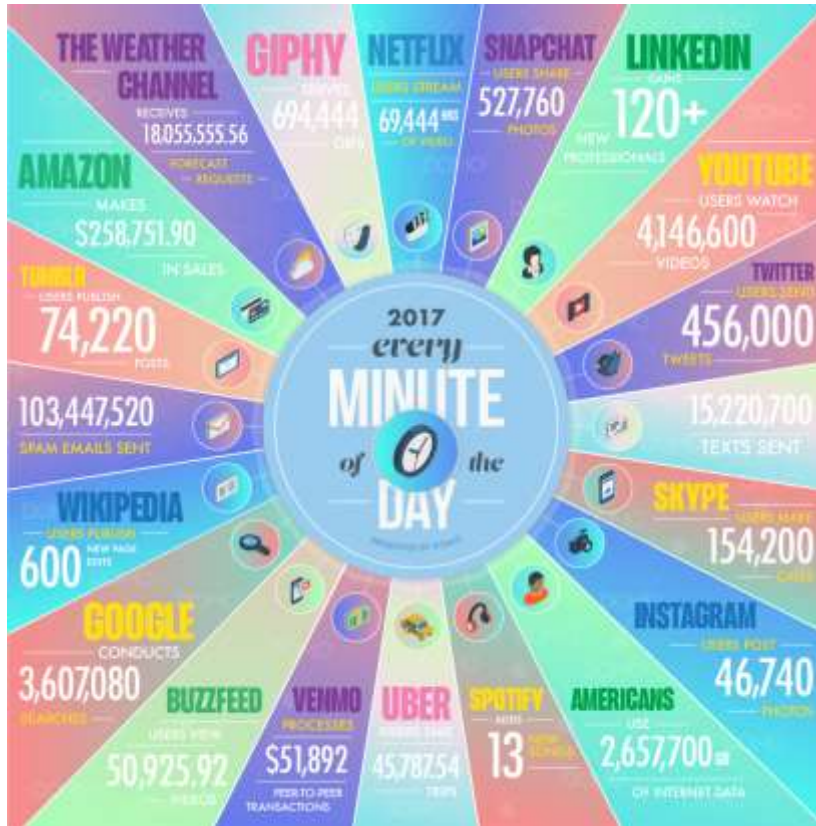
**100s of
millions
of GPS
enabled
devices** sold
annually

<http://www>

**2+
billion**
people on
the Web
by end
2011

The amount of data produced by us from the beginning of time till 2003 was 5 billion gigabytes. If you pile up the data in the form of disks it may fill an entire football field. The same amount was created in every two days in **2011**, and in every ten minutes in **2013**. This rate is still growing enormously.

Velocity



- If traditional data is like a drop of water, Big Data is like flowing river
- Speed at which data is generated by billions of devices, and communicated at the speed of light. (High speed Internet availability)
- Mobile devices can generate and communicate data from anywhere, at any time
- Processing should be faster than generation
- Analyse data while it is being generated without even putting it into databases

Image Source: <https://www.domo.com/blog/data-never-sleeps-5/>

Velocity (Speed)

- Data is being generated fast from different sources and need to be processed fast
- Online Data Analytics
- Late decisions → missing opportunities



- **Examples**

- **E-Promotions:** Based on your current location, your purchase history, what you like → send promotions right now for store next to you
- **Healthcare monitoring:** sensors monitoring your activities and body → any abnormal measurements require immediate reaction

Sources of Real-time/Fast Data



Social media and networks
(all of us are generating data)



Scientific instruments
(collecting all sorts of data)



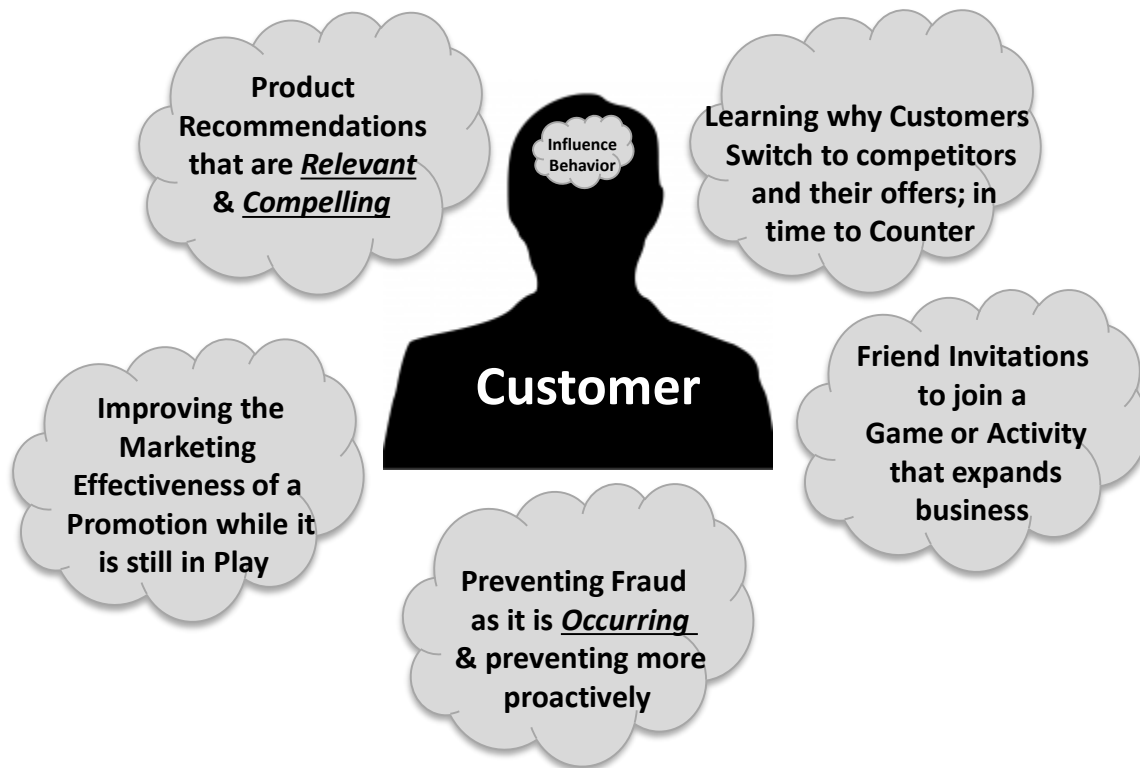
Mobile devices
(tracking all objects all the time)



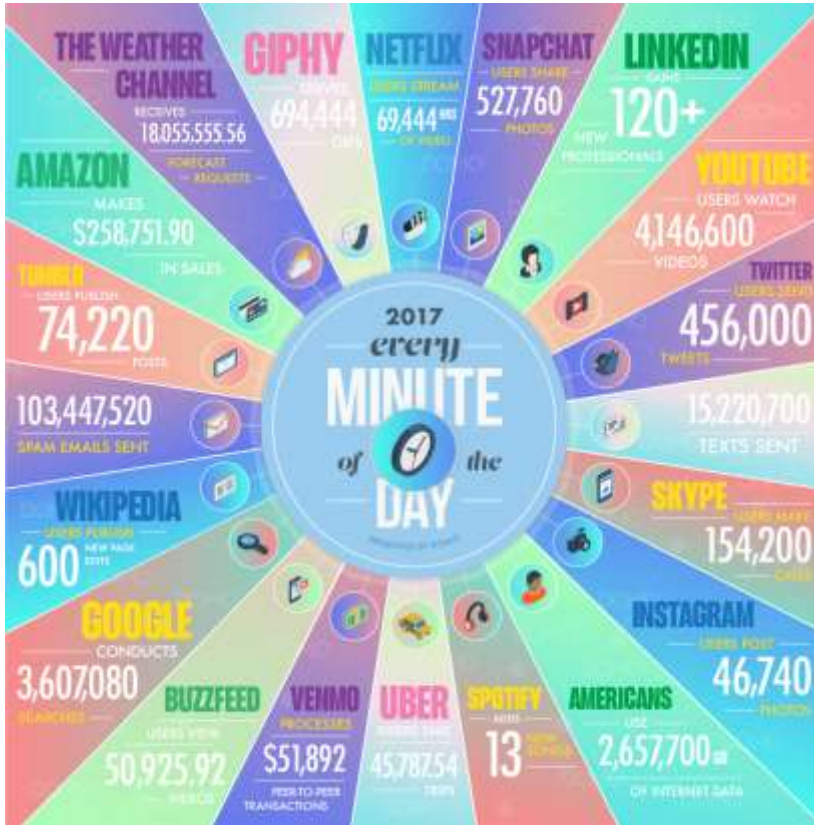
Sensor technology and networks
(measuring all kinds of data)

- The progress and innovation is no longer hindered by the ability to collect data
- But, by the ability to manage, analyze, summarize, visualize, and discover knowledge from the collected data in a timely manner and in a scalable fashion needs to be handled

Real-Time Analytics/Decision Requirement



Variety



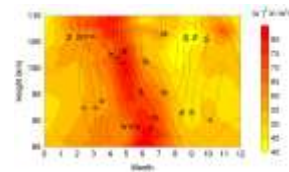
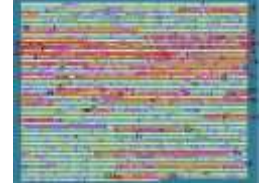
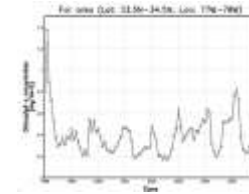
- Different types of data that we can use is inclusive of all forms of data, for all kinds of functions, from all sources and devices
- Generated by different entities
 - Humans
 - Machines (HW + SW)
 - Sensors

Variety (Complexity)

- Relational Data (Tables/Transaction/Legacy Data)
- Text Data (Web)
- Semi-structured Data (XML)
- Graph Data
 - Social Network, Semantic Web (RDF), ...
- Streaming Data
 - You can only scan the data once
- A single application can be generating/collecting many types of data ,
- Form Data type range in numbers to text, graph, map, audio, video and others
- Function: human conversation, songs and movies, business transaction records, machine operation performance data, new product design data, old archived data, etc.
- Source of data: Mobile phone and tablets, web access and search logs , Business transactional information, temperature and pressure sensors on machines, RFID tags on assets generate incessant and repetitive data.
- Big Public Data (online, weather, finance, detecting human faces from pictures, compare voice to identify the speaker, comparing handwritings to identify the writer, etc.)

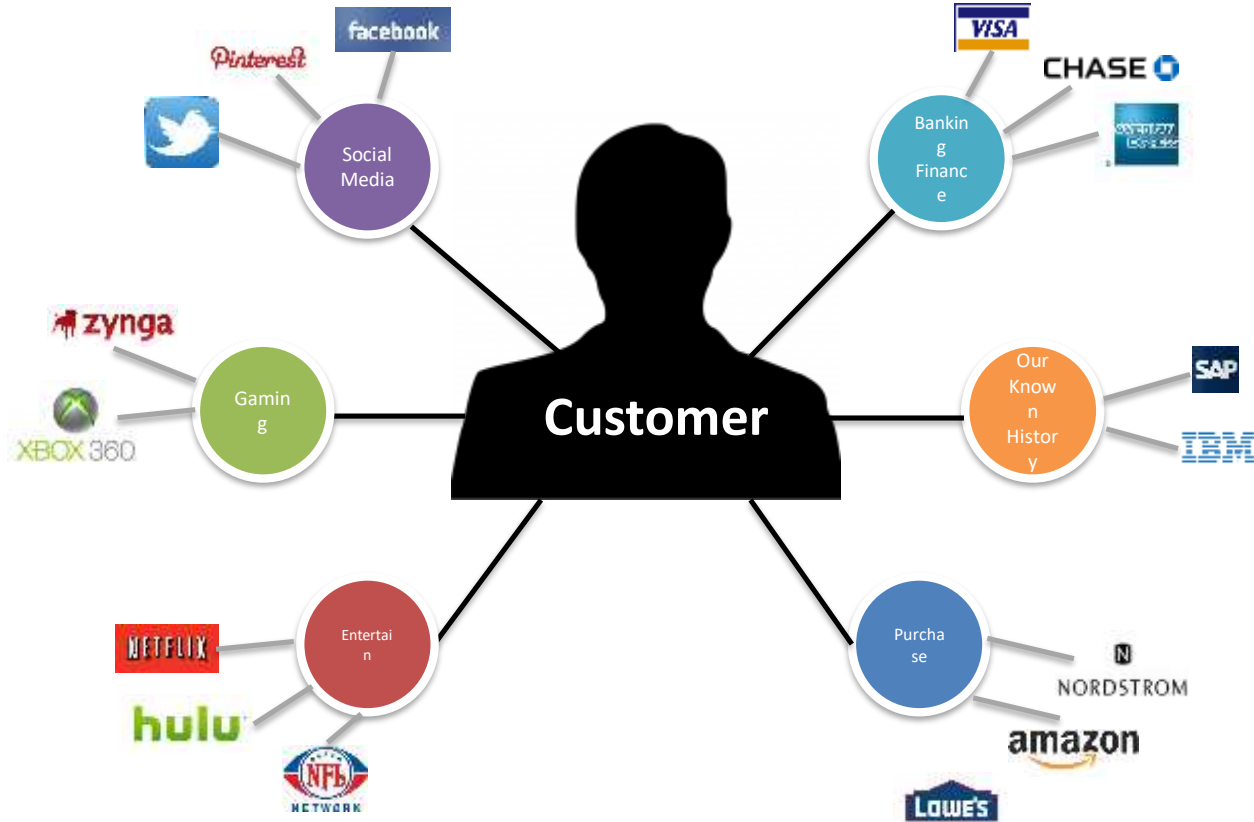
Broadly speaking there are three broad types of sources of data

Human to Human Communication, Human-Machine communication and
Machine to Machine communication.

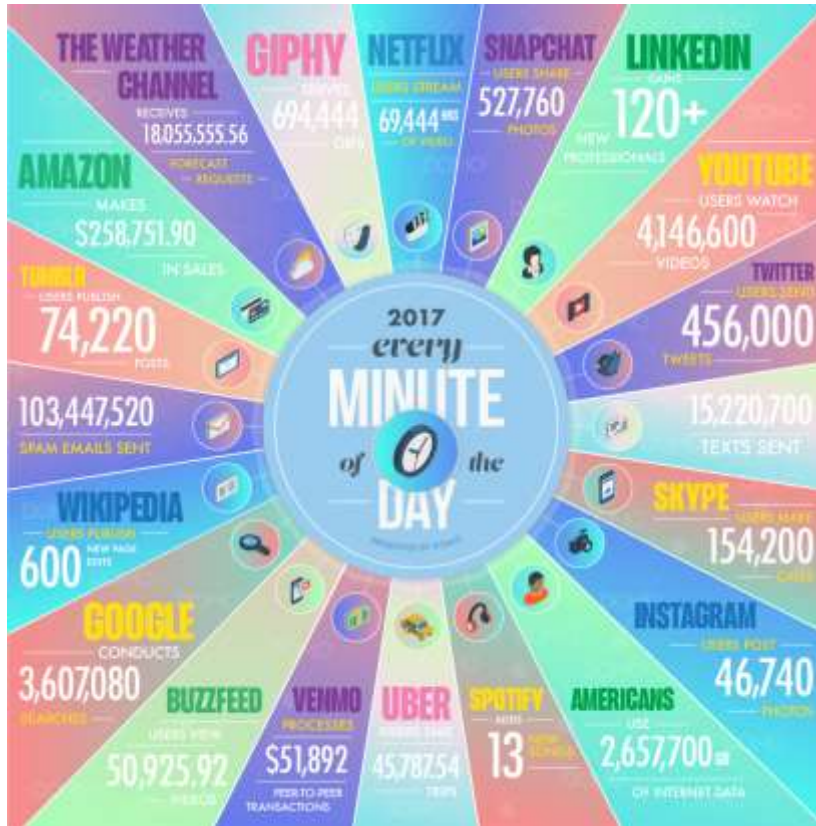


To extract knowledge → all these
types of data need to be linked together

A Single View to the Customer



Veracity

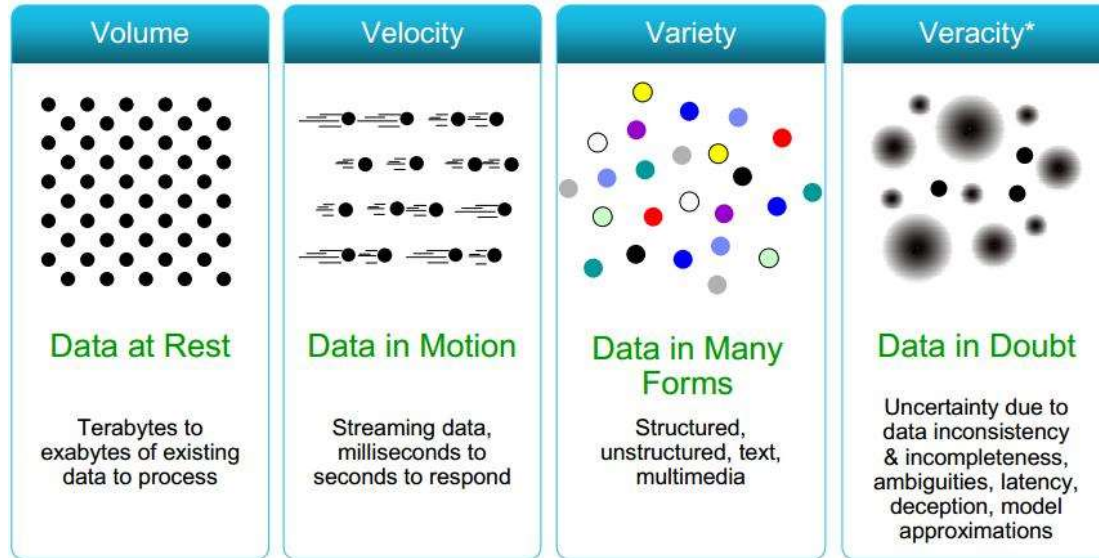


- Relates to the truthfulness, believability and quality of data
- Big Data is messy and there is considerable misinformation out there.
- The reasons for poor quality of data can range from technical error to human error to malicious intent.
- Volumes makes up for quality
 - Eg. Tweets with spelling mistakes, short words
 - u→you, thr→there, teh→the

Veracity

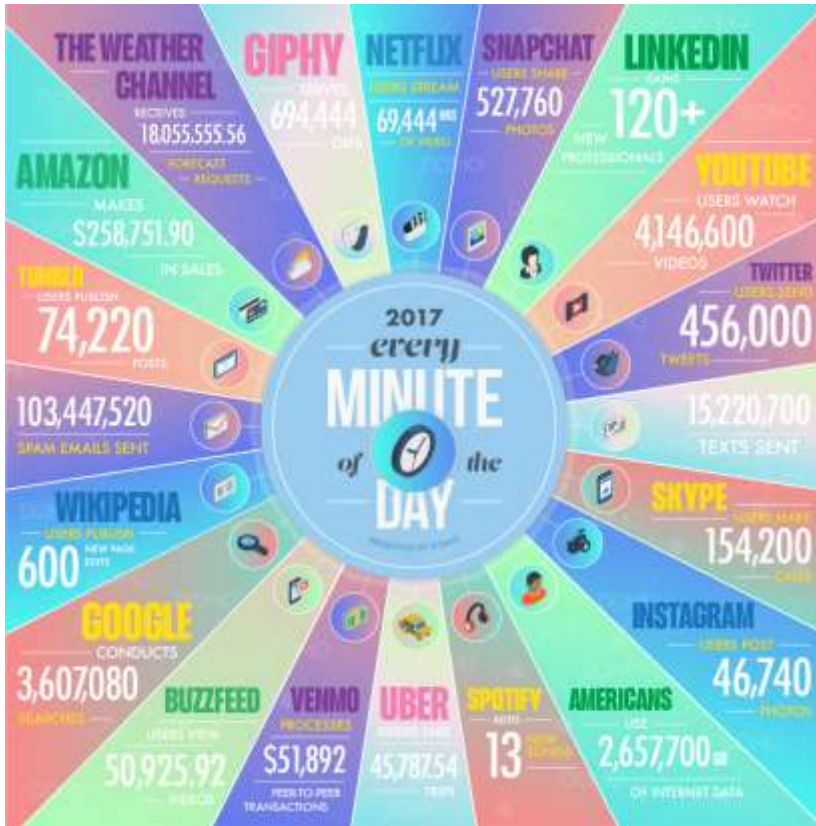
- The source of information may not be authoritative
 - Whitehouse.gov or nytimes.com is authentic and complete, but Wikipedia data may not be equally reliable.
- The data may not be communicated and received correctly because of human or technical failure
 - Sensors and communication machines may malfunction and may record and transmit incorrect data.
 - Urgency may require the transmission of the best data available at a point in time. Such data makes reconciliation with later, accurate, records more problematic.
- The data provided and received may, however also be intentionally wrong for competitive or security reasons.
 - There could be disinformation and malicious information spread for strategic reasons.

Some Make it 4V's



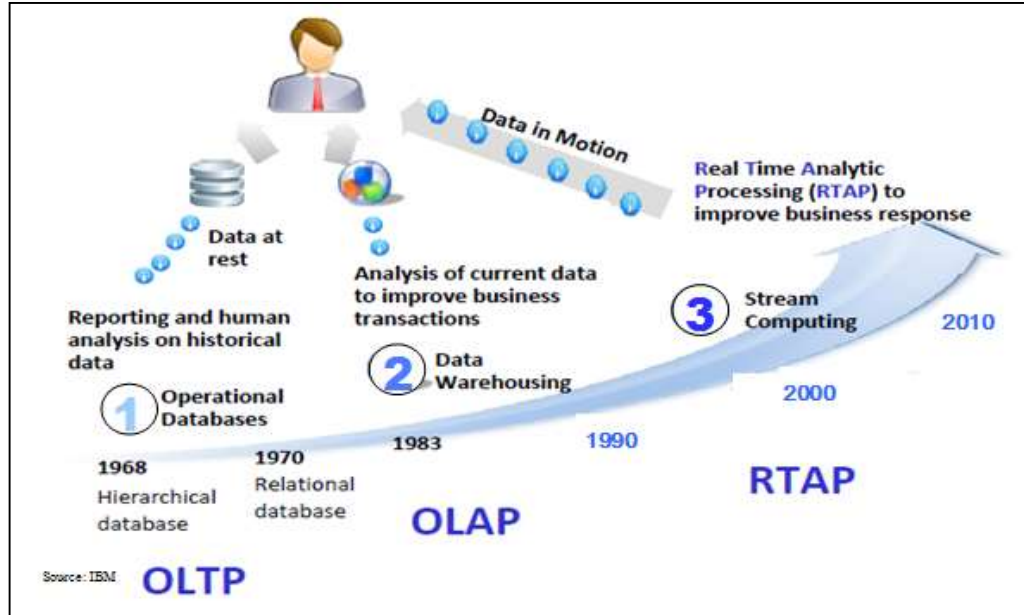
Additional V-s

Value



Getting value out of Big Data!!!

Harnessing Big Data



- **OLTP:** Online Transaction Processing (DBMSs)
- **OLAP:** Online Analytical Processing (Data Warehousing)
- **RTAP:** Real-Time Analytics Processing (Big Data Architecture & technology)

The Model Has Changed...

- **The Model of Generating/Consuming Data has Changed**

Old Model: Few companies are generating data, all others are consuming data



New Model: all of us are generating data, and all of us are consuming data

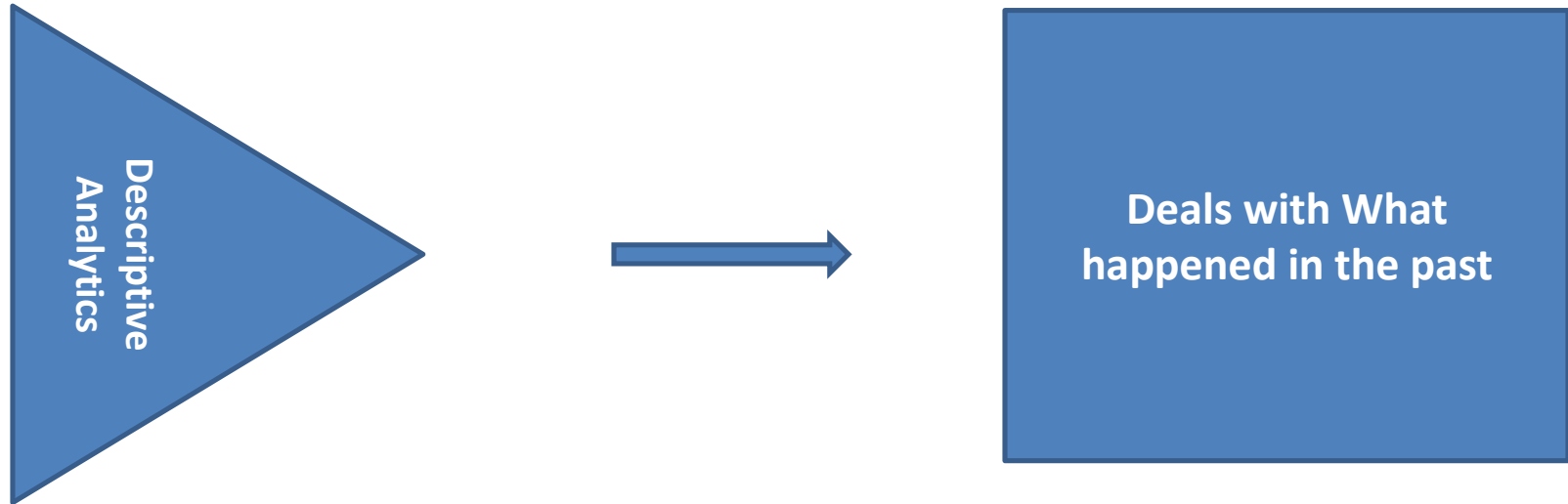


Types of Data Analytics

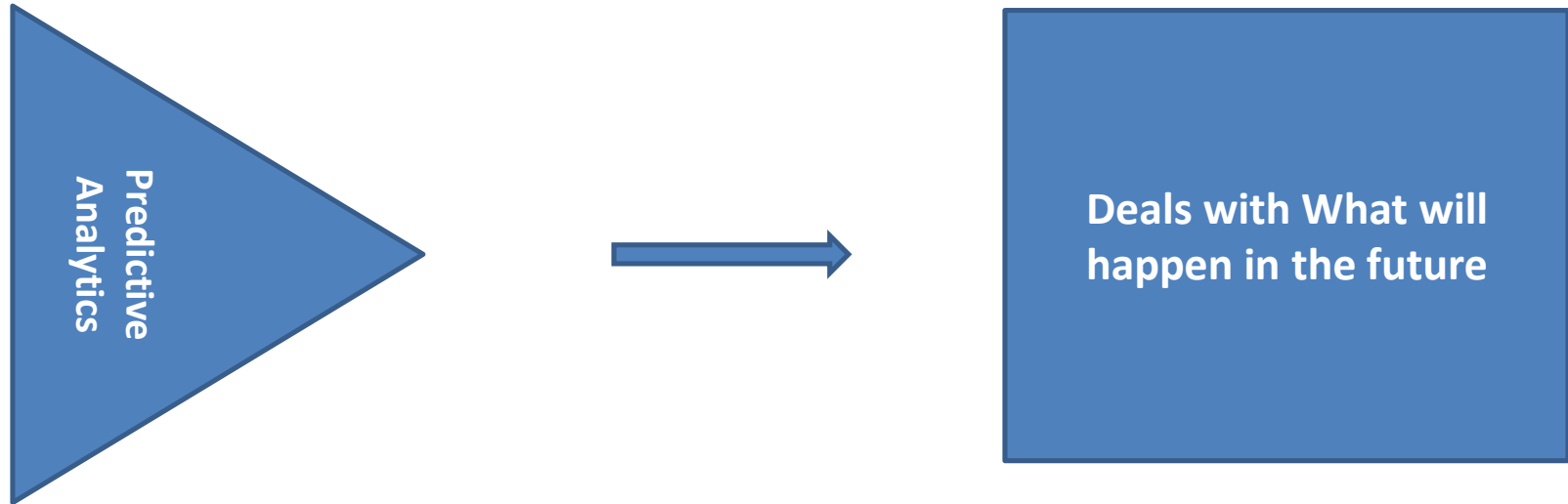
There are four major types of data analytics:

- 1. Descriptive (business intelligence and data mining)**
- 2. Predictive (forecasting)**
- 3. Prescriptive (optimization and simulation)**
- 4. Diagnostic analytics**

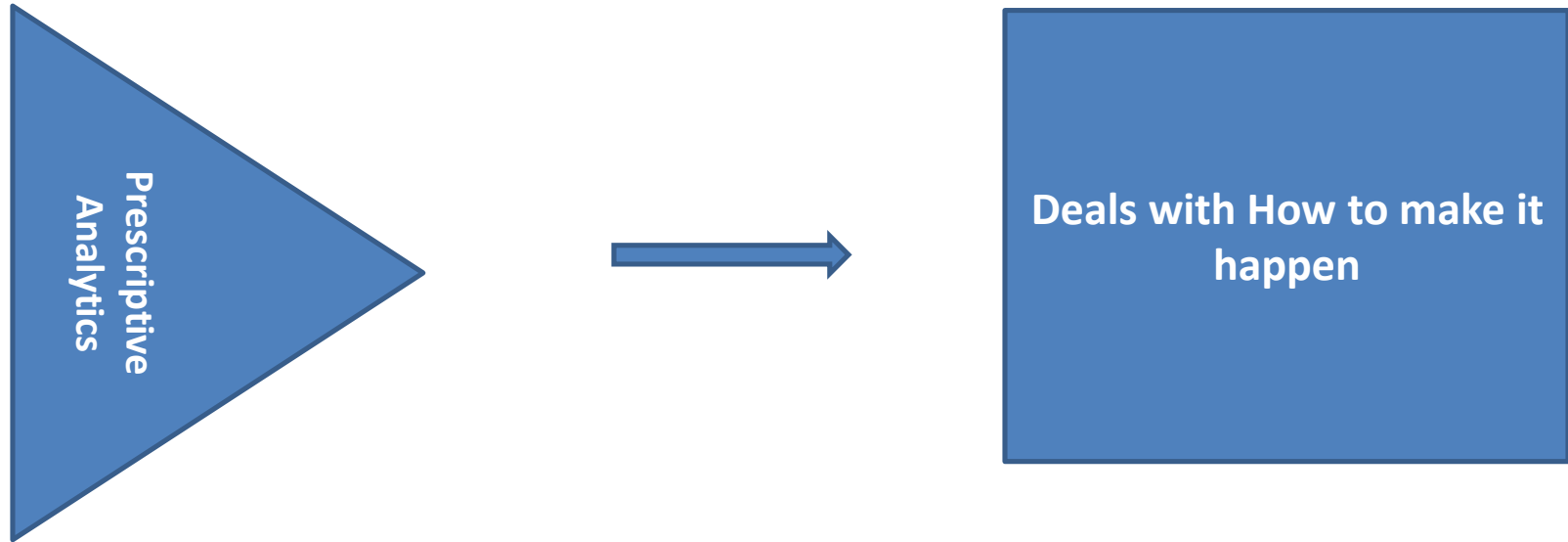
Descriptive analytics identifies many different relationships between customer and product.



Predictive analytics turn the data into valuable, actionable information



Prescriptive Analytics can suggest decision options on how to take advantage of a future opportunity or mitigate a future risk and illustrate the implication of each decision option.

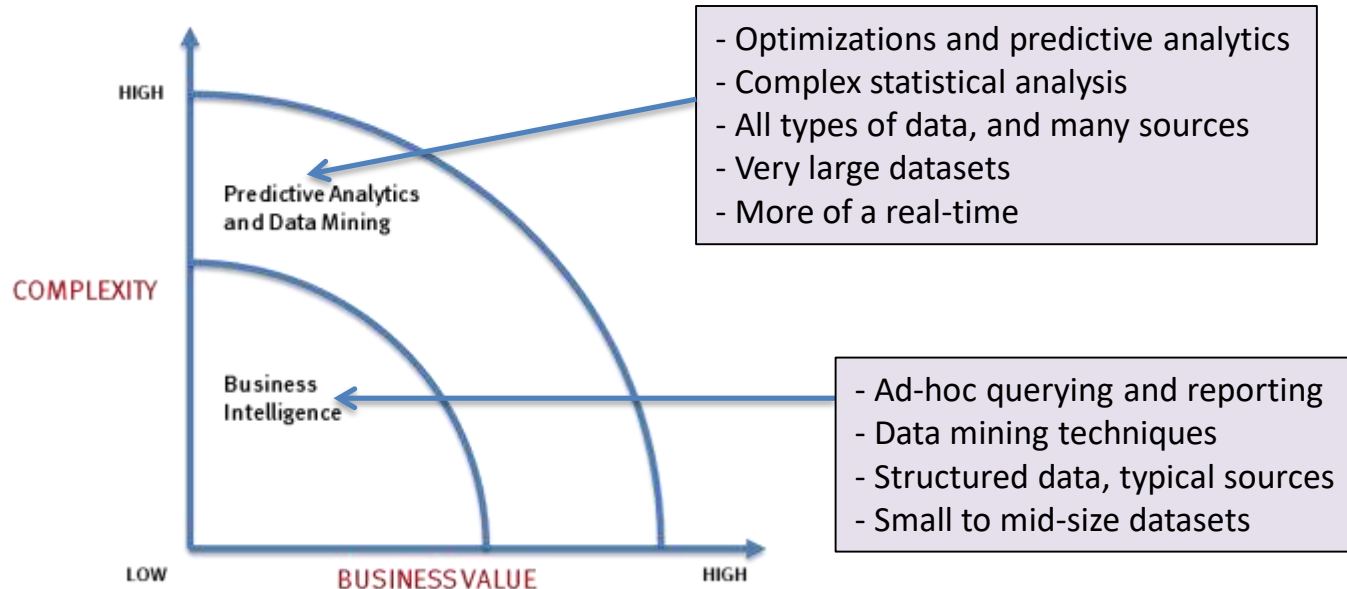


**Diagnostic
Analytics**



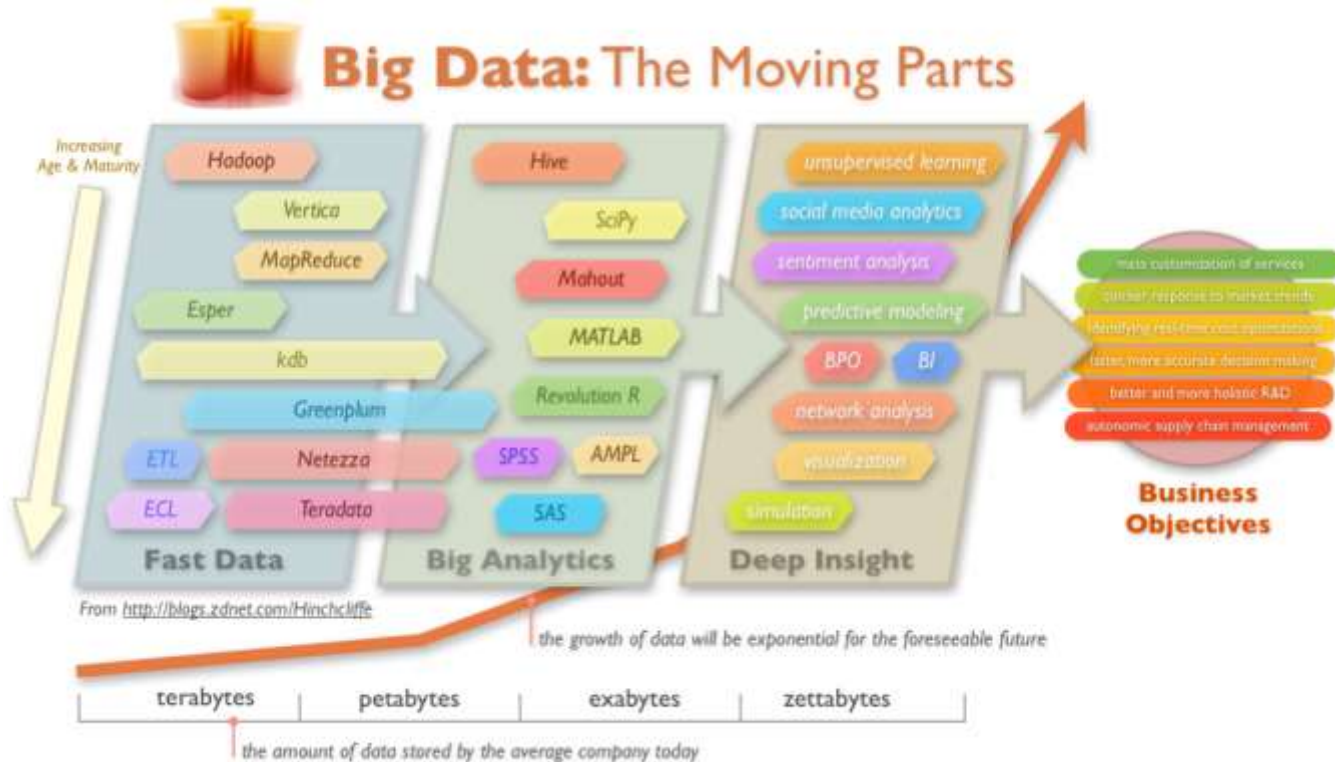
**Deals with why did it
happened in the past**

What's driving Big Data



Organizations that do not learn to engage with Big Data, could find themselves left far behind their competitors, landing in the bustbin of history.

Big Data Technology



Big Data Technologies

To harness the power of big data, you would require an infrastructure that can manage and process huge volumes of structured and unstructured data in real-time and can protect data privacy and security.

There are various technologies in the market from different vendors including Amazon, IBM, Microsoft, etc., to handle big data. While looking into the technologies that handle big data, we examine the following two classes of technology

- Operational and Analytical Systems

Operational Big Data

- Data collected and managed within information systems supporting operational work.
- This include systems like MongoDB that provide operational capabilities for real-time, interactive workloads where data is primarily captured and stored.
- NoSQL Big Data systems are designed to take advantage of new cloud computing architectures that have emerged over the past decade to allow massive computations to be run inexpensively and efficiently.

Analytical Big Data

These includes systems like Massively Parallel Processing (MPP) database systems and MapReduce that provide analytical capabilities for retrospective and complex analysis that may touch most or all of the data.

MapReduce provides a new method of analyzing data that is complementary to the capabilities provided by SQL, and a system based on MapReduce that can be scaled up from single servers to thousands of high and low end machines.

These two classes of technology are complementary and frequently deployed together.

Operational vs. Analytical Systems

	Operational	Analytical
Latency	1 ms - 100 ms	1 min - 100 min
Concurrency	1000 - 100,000	1 - 10
Access Pattern	Writes and Reads	Reads
Queries	Selective	Unselective
Data Scope	Operational	Retrospective
End User	Customer	Data Scientist
Technology	NoSQL	MapReduce, MPP Database

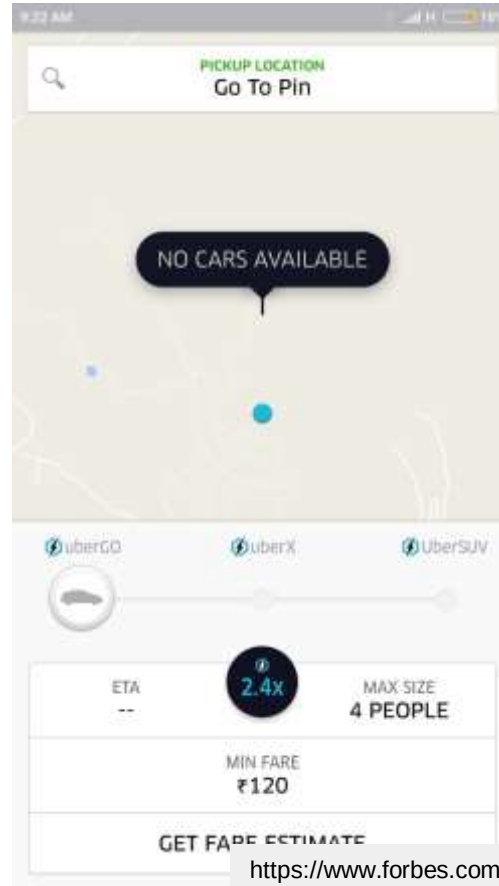
Use cases

Use Case: Big Data in Oil & Gas Drilling



<http://analytics-magazine.org/how-big-data-is-changing-the-oil-a-gas-industry/>

Use Case: Uber - Pay Surge Pricing if Battery is Low



Big Data Challenges

Big Data Challenges: Size does matter

1KB	Kilobyte
1MB	Megabyte
1GB	Gigabyte
1TB	Terabyte
1PB	Petabyte
1EB	Exabyte
1ZB	Zettabyte
1YB	Yottabyte

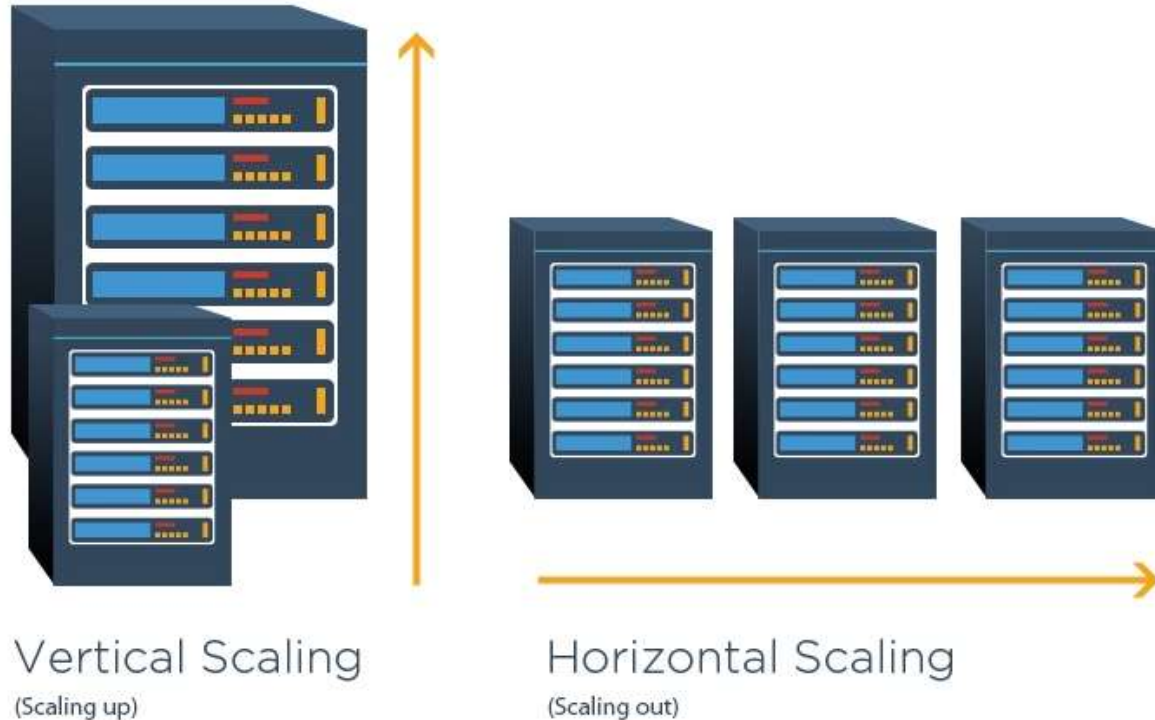
1 GB = 1 hr

1 TB = 1024 hrs = 102 days

1 PB = 286 yrs > **1 lifetime**

1 EB = 293K yrs

Big Data Challenges: Vertical Vs Horizontal Scaling



Big Data Challenges

Scaling



Source: <https://s-media-cache-ak0.pinimg.com/736x/10/0c/d0/100cd0da1c19e5d6f850ed23c3633714.jpg>

Big Data Challenges: Scale of infrastructure



Image Source: <https://datacenter.legrand.com>

Further Reading

- [A Brief History of Big Data Everyone Should Read](#)
- [Beyond Volume, Variety and Velocity is the Issue of Big Data Veracity](#)
- What is big data? - [OpenSource.com](#) & [O'Reilly](#)
- [Uber Use Case](#)
- [5 Big Data Use Cases To Watch](#)
- [Best Big Data Analytics Use Cases](#)
- [The 5 game changing big data use cases](#)
- [Big Data - The 5 Vs Everyone Must Know](#)
- [Top SlideShare Presentations on Big Data](#)
- [Google Data Center 360° Tour](#)

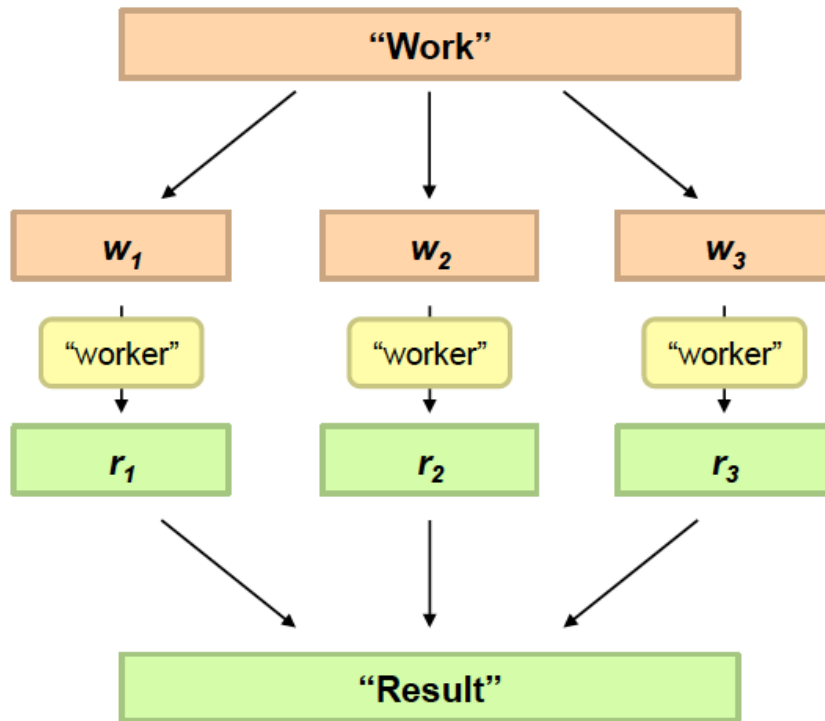
How to store *huge* files?

Requirements?

- Efficient Access
- Effective utilization of space
- Redundancy (Failsafe)
 - Given: probability of 1 disk failing is 1% per year
 - What are the chances that 1 out of 10^3 disk fails at a data center?

- Philosophy to Scale for Big Data?

Divide and Conquer



Divide Work



Combine Results

Distributed processing is non-trivial

- How to assign tasks to different workers in an efficient way?
- What happens if tasks fail?
- How do workers exchange results?
- How to synchronize distributed tasks allocated to different workers?



Big data storage is challenging

- Data Volumes are massive
- Reliability of Storing PBs of data is challenging
- All kinds of failures: Disk/Hardware/Network Failures
- Probability of failures simply increase with the number of machines ...



One popular solution: Hadoop



Hadoop Cluster at Yahoo! (Credit: Yahoo)

.Hadoop

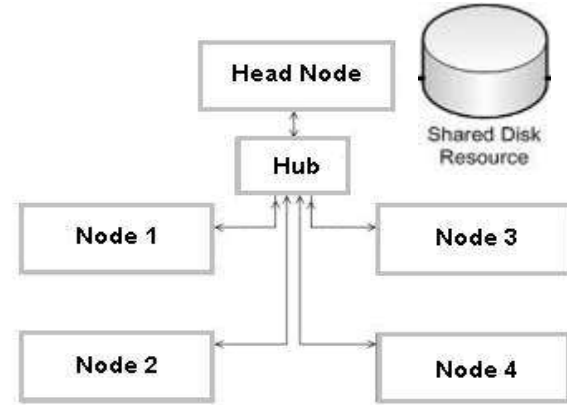
- Framework that allows for the distributed processing of large data sets
 - across clusters of computers
 - using simple programming models.
- Designed to scale up from single servers to thousands of machines, each offering local computation and storage.
- Designed to detect and handle failures at the application layer
 - delivering a highly-available service on top of a cluster of computers, each of which may be prone to failures.

What is Hadoop?

- Hadoop is a framework that uses distributed storage and parallel processing to store and manage big data.
- It is the software most used by data analysts to handle big data
- There are three components of Hadoop:
 1. Hadoop HDFS - Hadoop Distributed File System (HDFS) is the storage unit.
 2. Hadoop MapReduce - Hadoop MapReduce is the processing unit.
 3. Hadoop YARN - Yet Another Resource Negotiator (YARN) is a resource management unit.
 4. Hadoop Common - The common utilities that support the other Hadoop modules.

Motivation - Traditional Distributed systems

- ❑ Processor Bound
- ❑ Using multiple machines
- ❑ Developer is burdened with managing too many things
 - Synchronization
 - Failures
- ❑ Data moves from shared disk to compute node
- ❑ Cost of maintaining clusters
- ❑ Scalability as and when required not present



What we need



Handling failure

- One computer = fails once in 1000 days
- 1000 computers = 1 per day



Petabytes of data to be processed in parallel

- 1 HDD= 100 MB/sec
- 1000 HDD= 100 GB/sec



Easy scalability

- Relative increase/decrease of performance depending on increase/decrease of nodes

Hadoop: Myth Vs Truth

Myth	Truth
HDFS is a database	HDFS is a Distributed File System
Hadoop is a replacement of database warehouse	Compliments it, not a substitute
Hadoop is a complete, single product	Ecosystem , not just a product. HDFS and MapReduce being the key components
Hadoop is used only for unstructured data, web analytics	Enables many types of analytics

Who is using Hadoop



Also see

<https://www.dezyre.com/article/top-10-industries-using-big-data-and-121-companies-who-hire-hadoop-developers/69>

Hadoop offers

- Redundant, Fault-tolerant data storage
- Parallel computation framework
- Job coordination



Programmers

*No longer need to
worry about*



**Q: Where file is
located?**

**Q: How to handle
failures & data
lost?**

**Q: How to divide
computation?**

**Q: How to
program for
scaling?**

A real world example of New York Times

- **Goal:** Make entire archive of articles available online: 11 million, from 1851
- **Task:** Translate 4 TB TIFF images to PDF files
- **Solution:** Used Amazon Elastic Compute Cloud (EC2) and Simple Storage System (S3)
- **Time:** ?
- **Costs:** ?



The
New York
Times

A real world example of New York Times

- **Goal:** Make entire archive of articles available online: 11 million, from 1851
- **Task:** Translate 4 TB TIFF images to PDF files
- **Solution:** Used Amazon Elastic Compute Cloud (EC2) and Simple Storage System (S3)
- **Time:** < 24 hours
- **Costs:** \$240

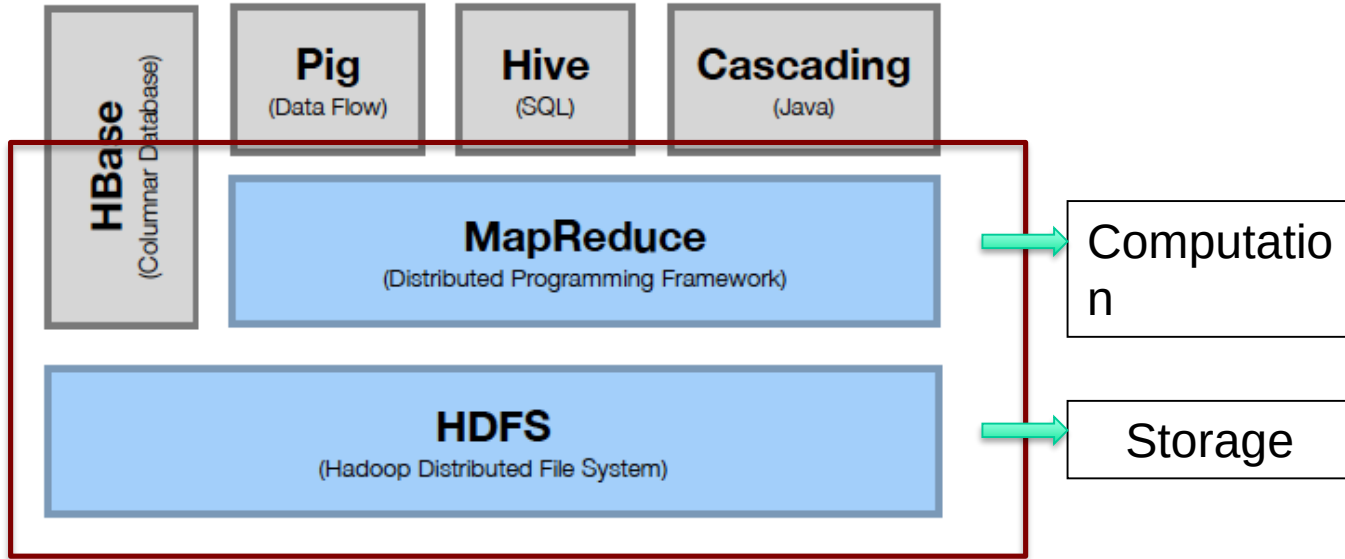
The logo of The New York Times, featuring the words "The", "New York", and "Times" stacked vertically in a black, bold, serif font.

A little history on Hadoop

- Hadoop was started with **Doug Cutting and Mike Cafarella** in the year 2002 when they both started to work on Apache Nutch project.
- Apache Nutch project was the process of building a search engine system that can index 1 billion pages.
- The system will cost around half a million dollars in hardware, and along with a monthly running cost of \$30, 000 approximately.
- **In 2003**, they read white paper that described the architecture of Google's distributed file system, called **GFS (Google File System)** published by Google, for storing the large data sets.
- **In 2004**, Google published **MapReduce**.
- **In 2006**, Doug Cutting joined Yahoo along with Nutch project.
- **In 2007**, Yahoo successfully tested Hadoop on a 1000 node cluster and start using it.
- **In 2008**, Yahoo released Hadoop as an open source project to ASF(Apache Software Foundation). And in **July of 2008**, Apache Software Foundation **successfully tested a 4000 node cluster with Hadoop**.
- **In 2009**, Hadoop was successfully tested to sort a PB (PetaByte) of data in less than 17 hours for handling billions of searches and indexing millions of web pages.



Hadoop Stack



Motivation Questions

- **Problem 1:** Data is too big to store on one machine.
- **HDFS:** Store the data on multiple machines!

Motivation Questions

- **Problem 2:** Very high end machines are too expensive
- **HDFS:** Run on commodity hardware!

Motivation Questions

- **Problem 3:** Commodity hardware will fail!
- **HDFS:** Software is intelligent enough to handle hardware failure!

Motivation Questions

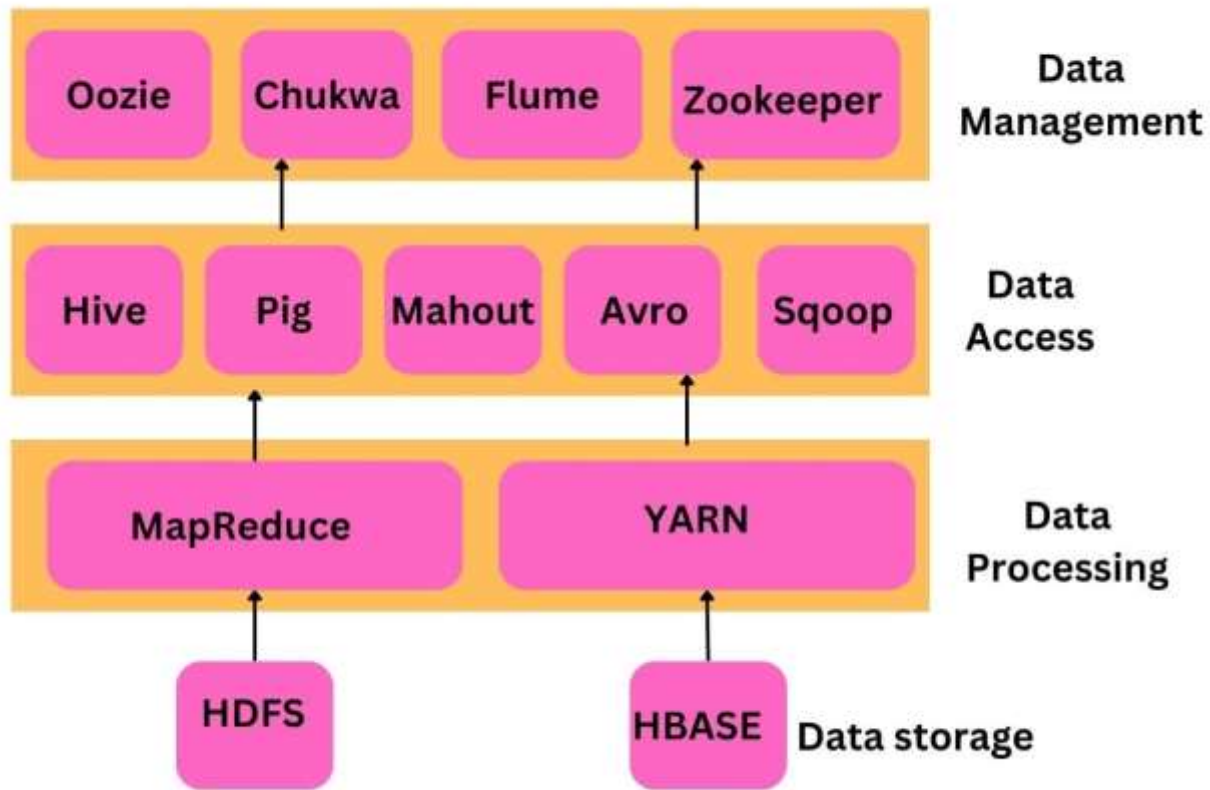
- **Problem 4:** What happens to the data if the machine stores the data fails?
- **HDFS:** Replicate the data!

Motivation Questions

- **Problem 5:** How can distributed machines organize the data in a coordinated way?
- **HDFS:** Master-Slave Architecture!

- There are *four major elements of Hadoop* i.e. **HDFS, MapReduce, YARN, and Hadoop Common**.
- Following are the components that collectively form a Hadoop ecosystem:
 - HDFS: Hadoop Distributed File System
 - YARN: Yet Another Resource Negotiator
 - MapReduce: Programming based Data Processing
 - PIG, HIVE: Query based processing of data services
 - HBase: NoSQL Database
 - Mahout, Spark MLlib: Machine Learning algorithm libraries
 - Solar, Lucene: Searching and Indexing
 - Zookeeper: Managing cluster
 - Oozie: Job Scheduling

Hadoop Ecosystem



HDFS

Hadoop distributed File System

HDFS

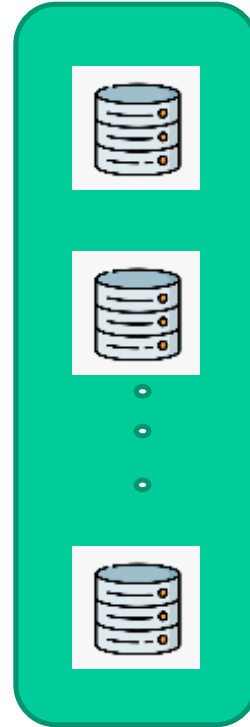
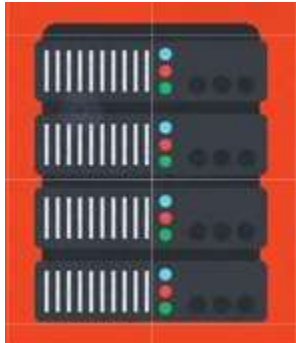
- Data storage system used by Hadoop
 - Hadoop: Project to develop open-source software for reliable, scalable, distributed computing★
- Components
- Architecture
- Tasks / Services

Hadoop

Hadoop framework allow us to store large volume of data at multiple location

Hadoop framework also allow us to process data in parallel manner

HDFS



- ❑ Hadoop Distributed File System (HDFS) is a specially designed file system for storing huge dataset in commodity hardware.

HDFS has two main components

Data node

Name node

Components of HDFS



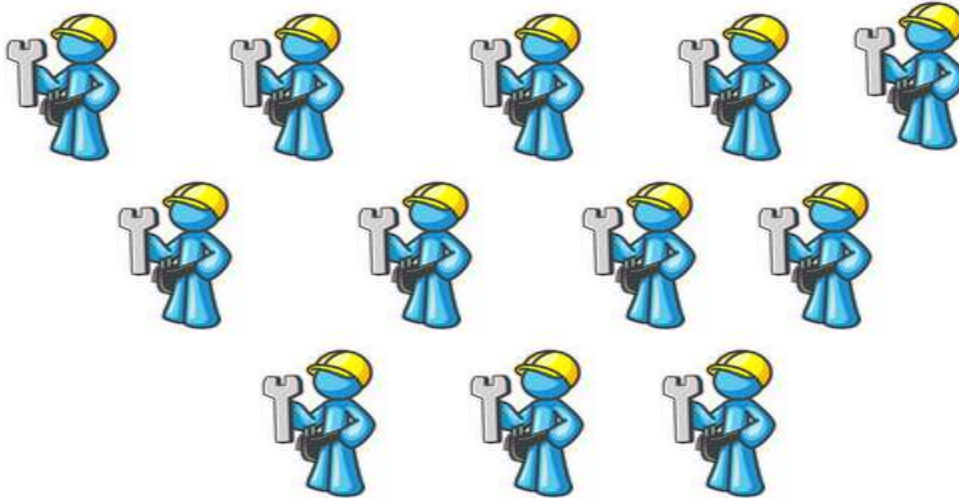
Secondary NameNode



Active NameNode



Standby NameNode



DataNodes

Name Node

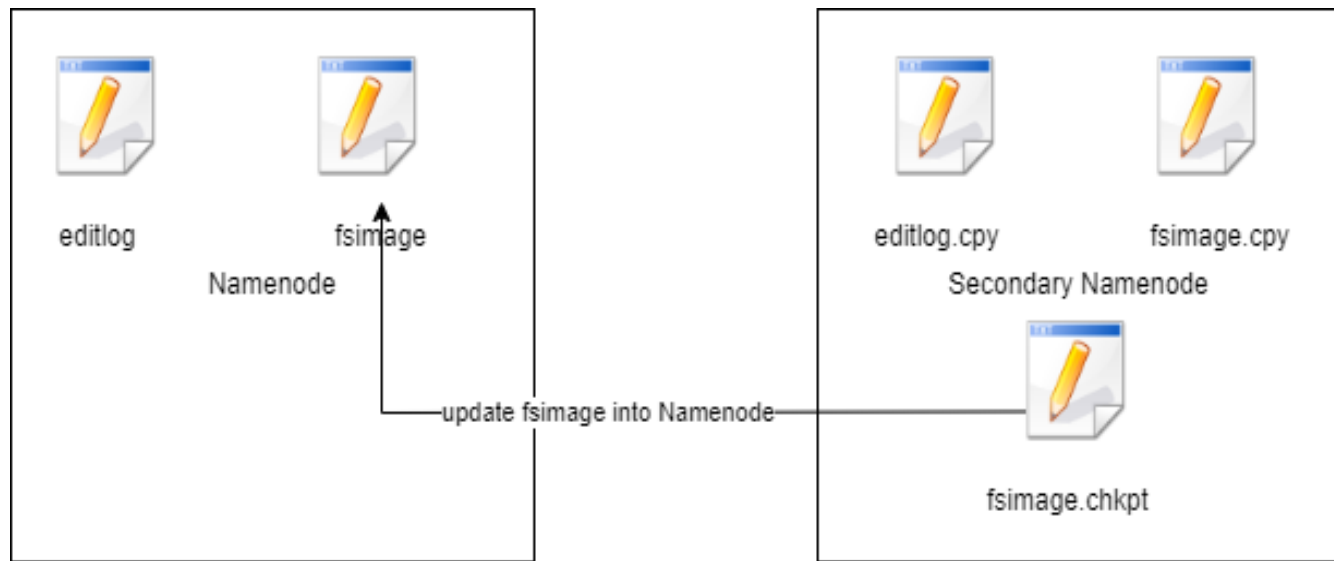
- ❑ The NameNode is a critical component of HDFS and serves as the master server.
- ❑ It is responsible for managing the metadata of the file system.
- ❑ This metadata includes information about the structure of the file system, the location of data blocks, and the health of DataNodes.

Name Node

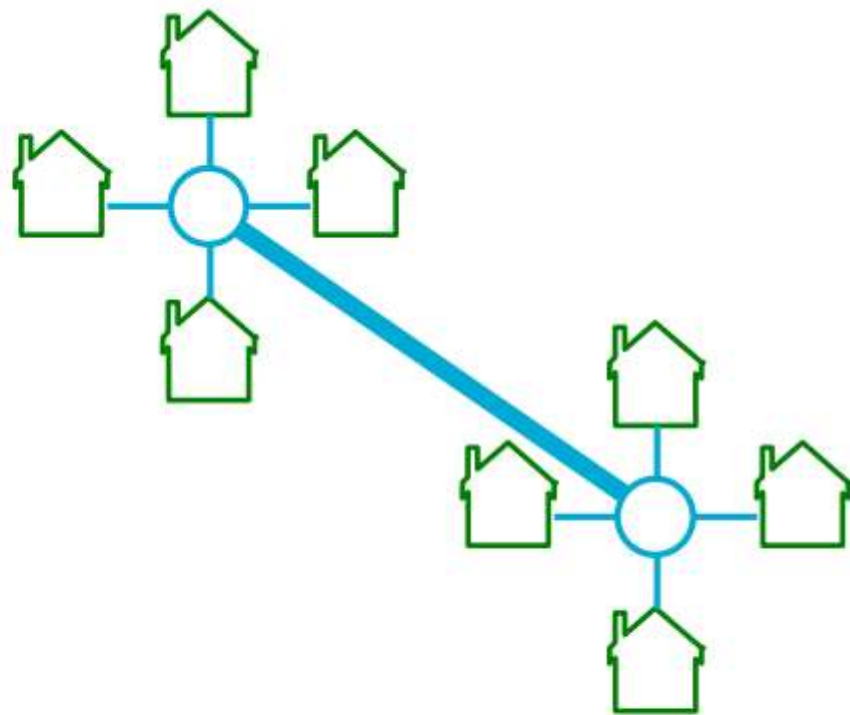
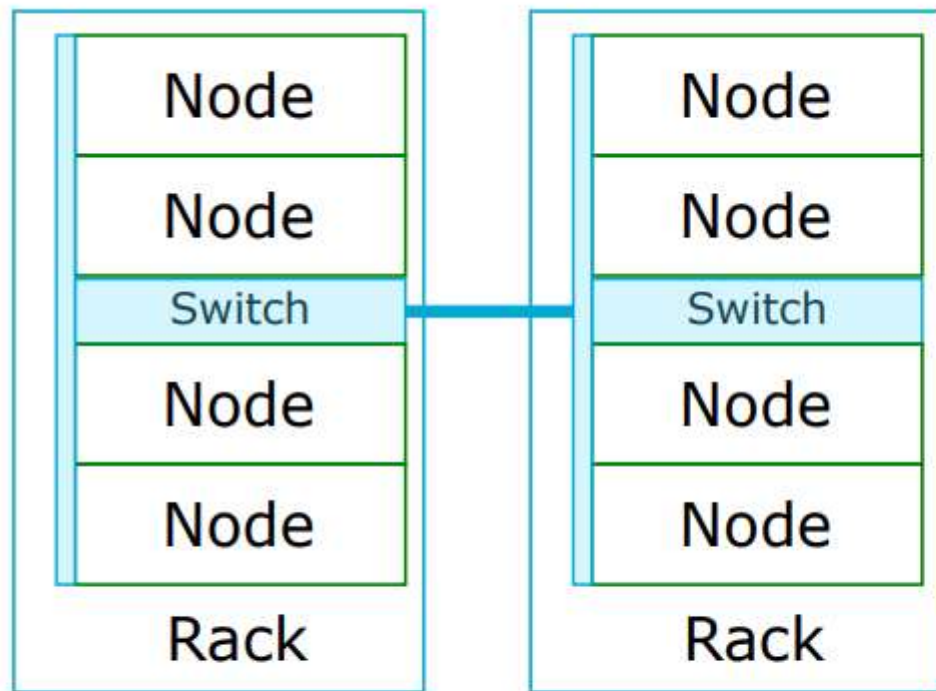
- ❑ The NameNode does not store the actual data content of files but keeps track of the metadata, such as file names, permissions, and the hierarchy of directories.
- ❑ It maintains an image of the entire file system namespace and the mapping of data blocks to DataNodes.

- ❑ Editlog : keep track of recent changes
- ❑ Fsimage : Keep track of every changes made on HDFS since the beginning

Secondary Namenode



- ❑ Racks
- ❑ Switch
- ❑ Cluster

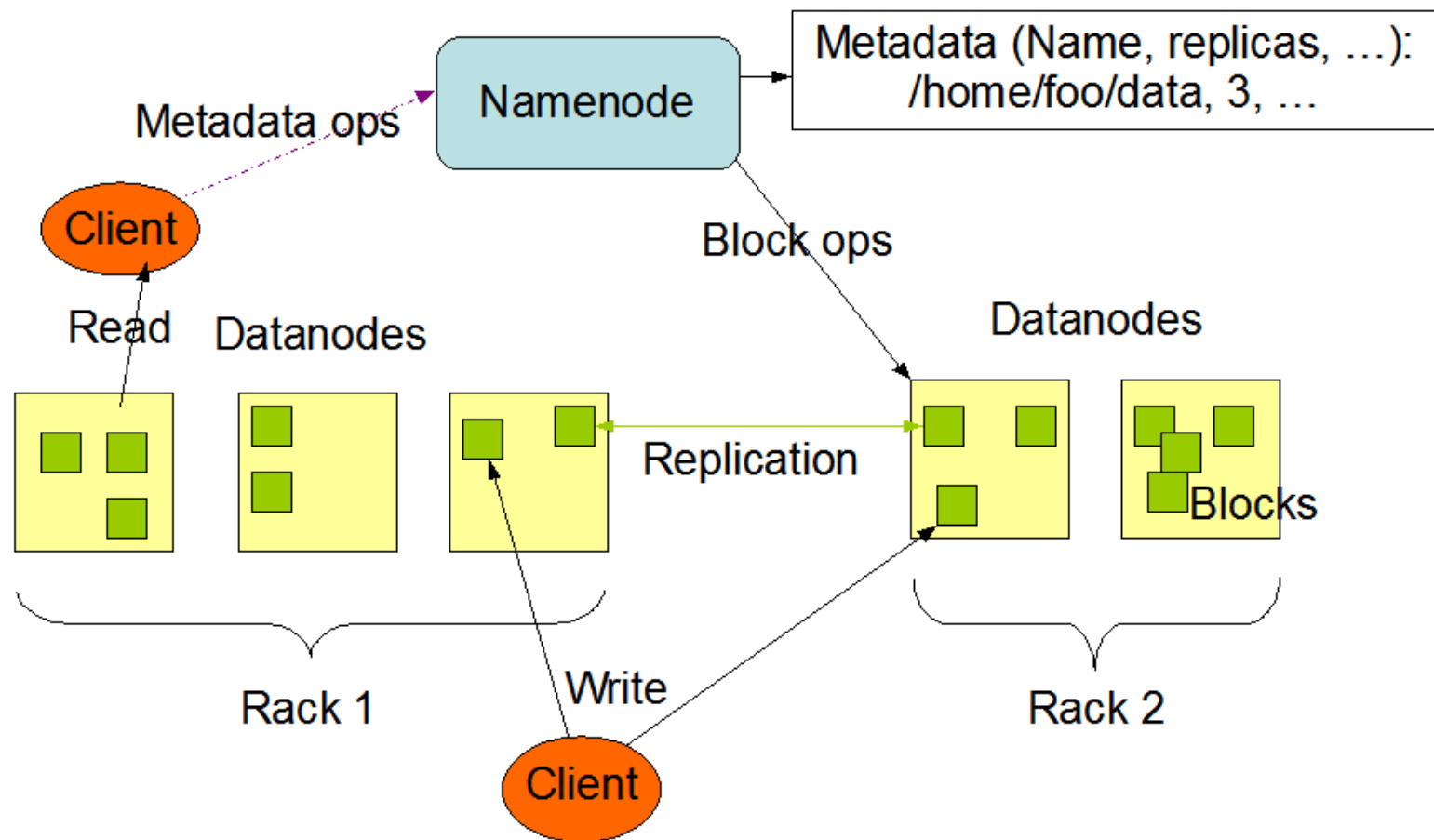


Data Node

- ❑ DataNodes works as a *Slave*
- ❑ DataNodes are mainly utilized for storing the data in a Hadoop cluster
- ❑ The number of DataNodes can be from 1 to 500 or even more than that, the more number of DataNode your Hadoop cluster has More Data can be stored.

- ❑ A file is split into one or more blocks and these blocks are stored in a set of DataNodes.
- ❑ The NameNode executes file system namespace operations like opening, closing, and renaming files and directories. It also determines the mapping of blocks to DataNodes.
- ❑ The DataNodes are responsible for serving read and write requests from the file system's clients.
- ❑ The DataNodes also perform block creation, deletion, and replication upon instruction from the NameNode.

HDFS Architecture

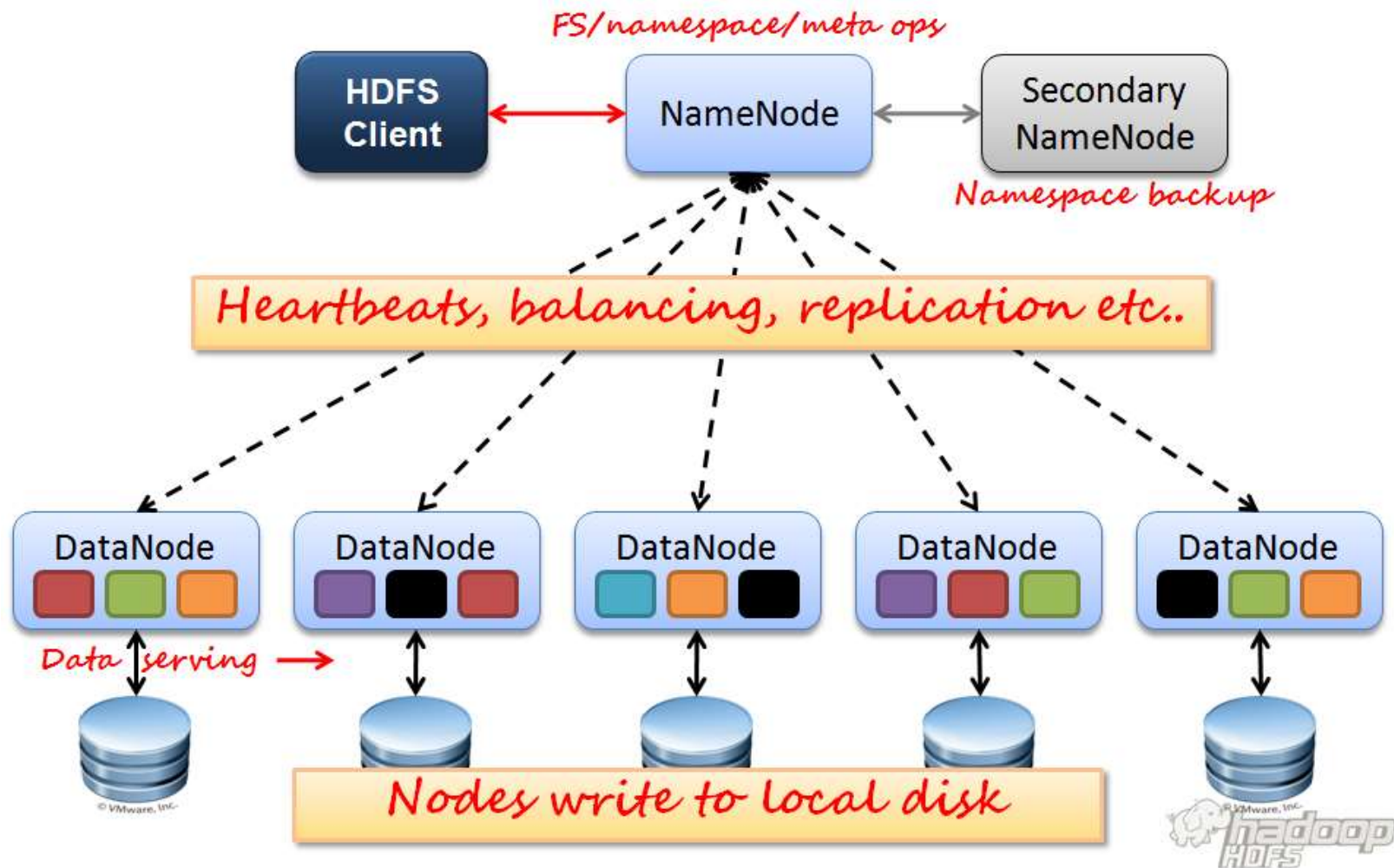


- ❑ Hadoop Distributed File System (HDFS) is the open source version of the Google File System (GFS) Allows reliable and efficient storage and access of large write-once, read-many (WORM) files
- ❑ NameNode acts as a server that manages the filesystem
- ❑ DataNodes store data blocks and serve read/write requests
- ❑ Blocks are replicated to allow for fault tolerance and fast reads

DataNodes

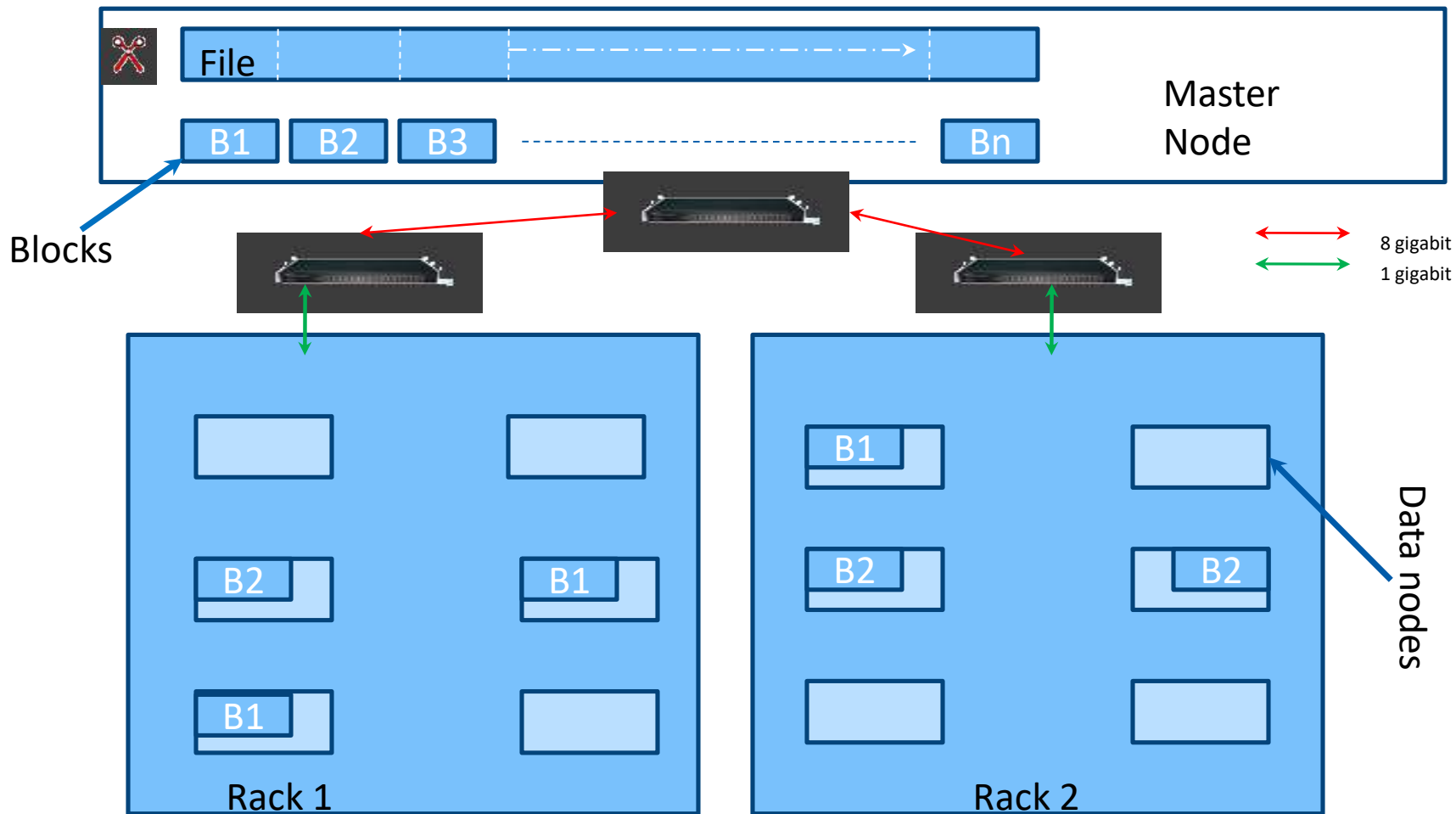
- ❑ Block report
- ❑ Heartbeats

Architecture



Storing file on HDFS

- Default replication factor is 3
 - first replica of a block will be stored on a local rack
 - the next replica will be stored on a remote rack
 - the third replica will be stored on the same remote rack but on a different Datanode
- More replicas?
 - the rest will be placed on random Datanodes
 - As far as possible, no more than two replicas are kept on the same rack



Tasks of NameNode

❏ Manages File System

- mapping files to blocks and blocks to data nodes

❏ Maintaining status of data nodes

➤ Heartbeat

- Datanode sends heartbeat at regular intervals
- If heartbeat is not received, datanode is declared dead

➤ Blockreport

- DataNode sends list of blocks on it
- Used to check health of HDFS

- ❑ The NameNode does not directly call DataNodes. It uses replies to heartbeats to send instructions to the DataNodes. The instructions include commands to:
 - ❑ replicate blocks to other nodes;
 - ❑ remove local block replicas;
 - ❑ re-register or to shut down the node;
 - ❑ send an immediate block report

NameNode Functions

- ❑ Replication
 - On Datanode failure
 - On Disk failure
 - On Block corruption
- ❑ Data integrity
 - Checksum for each block
 - Stored in hidden file
- ❑ Rebalancing - balancer tool
 - Addition of new nodes
 - Decommissioning
 - Deletion of some files

- ❑ The primary objective of HDFS is to store data reliably in the presence of failures
- ❑ Three common failures that it must handle are:
 - DataNode failure
 - NameNode failure
 - Network partition

Data integrity

What if a block of data fetched from a DataNode arrives corruptedFault in a storage device

- Network faults
- Buggy software

An HDFS client creates a checksum for every block of its file and stores it in the HDFS namespace

- When the client retrieves the contents of a file it verifies that they match...if not it must retrieve the block from another replica

Cluster Rebalancing

- ❑ HDFS architecture is compatible with data rebalancing schemes
- ❑ A scheme may automatically move data from one DataNode to another if the free space on a DataNode is falling below a certain threshold

Replication -Pipeline

- ❑ When a client is writing data to an HDFS file with a replication factor of three, the NameNode retrieves a list of DataNodes using a replication target choosing algorithm.
- ❑ It contains the DataNodes that will host a replica of that block.
- ❑ The client then writes to the first DataNode, starts receiving the data in portions, writes each portion to the second DataNode in the list.
- ❑ The second DataNode, in turn starts receiving each portion of the data block, writes that portion to its repository and then flushes that portion to the third DataNode.
- ❑ Finally, the third DataNode writes the data to its local repository.
- ❑ Thus, the data is pipelined from one DataNode to the next.

HDFS Robustness

❑ Safemode

- At startup: No replication possible
- Receives Heartbeats and Blockreports from Datanodes
- Only a percentage of blocks are checked for defined replication factor

❑ Replicate blocks wherever necessary

All is well 😊 → Exit Safemode

HDFS Summary

- ❑ Fault tolerant
- ❑ Scalable
- ❑ Reliable
- ❑ File are distributed in large blocks for
 - Efficient reads
 - Parallel access

Questions?

